

Isabelle Laing

CONTACT	Student Email: isabelle.laing@mail.utoronto.ca Phone: +1 (416) 771-1737 GitHub: isabelle-laing Website: isabelle-laing.github.io
EDUCATION	University of Toronto Expected Graduation in June 2027 Honours Bachelor of Science at the University of Toronto • Astronomy & Physics Specialist • Mathematics Minor
AWARDS	Best Talk Award at the Canadian Astronomical Society’s Annual General Meeting, 2026 Data Science & AI Talent Showcase Poster Award, University of Toronto, 2026 Samuel Beatty Fund Travel Award, Physics Department University of Toronto, 2026 SURP Poster Award, University of Toronto, 2025 University of Toronto Admissions Scholarship, 2023
RESEARCH EXPERIENCE	SURF – Summer Undergraduate Research Fellowship May 2026 – Present Canadian Institute for Theoretical Astrophysics (CITA), University of Toronto • Supervised by Dr. Aviad Levis and Dr. Rohan Dahale. • Building a pipeline in Python/JAX to model polarised light curves from plasma hotspots orbiting Sgr A*. AST430 – Student Research Course September 2025 – April 2026 David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto • Supervised by Dr. Renée Hložek, on the Hnauts research team. • Generated a full pipeline for stream simulation and investigated how different dark matter models affect the streams themselves. Student Research Internship May 2025 – August 2025 Sydney Institute for Astrophysics, University of Sydney • Supervised by Dr. Joss Bland-Hawthorn and Dr. Thorsten Tepper-Garcia. • Identified and analysed clumps in galactic simulations for the Galacticos Group. SURP – Summer Undergraduate Research Program May 2024 – August 2024 David A. Dunlap Department of Astronomy & Astrophysics, University of Toronto • Supervised by Dr. Josh Speagle and Dr. Ting Li. • Created a machine learning pipeline to derive stellar parameters and chemical abundances from DESI stellar spectra. • Focus on galactic archaeology, tracing the assembly history of the Milky Way halo through its surviving stars. • Released a catalogue containing neural-network-derived stellar parameters and chemical abundances for over 12 million Milky Way Halo stars. • This project continued as a paid student research internship with the Department of Statistical Sciences (UofT) from September 2024 to April 2025, and then as a student research course, AST430, from May 2025 to September 2025.

- PRESENTATIONS**
- **Poster & Lightning Talk**, *STATSTRO 2026, July 2026*
 - **Talk**, *Canadian Astronomical Society's Annual General Meeting, June 2026*
 - **Presentation**, *University of Toronto Physics Union Undergraduate Research Conference, March 2026*
 - **Presentation**, *Canadian Conference for Undergraduate Women & Gender Minorities in Physics, January 2026*
 - **Poster**, *Data Sciences Institute Talent Showcase, January 2026*
 - **Poster**, *Vector Foundation Models for Science Workshop, November 2025*
 - **Poster**, *University of Toronto Summer Undergraduate Research Program Poster Fair, August 2024 and August 2025*
 - **Poster**, *University of Toronto Research Opportunity Program (ROP) Poster Fair, April 2025*
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- PROFESSIONAL SKILLS**
- Languages**
- Native Speaker in French and English
 - Bi Spanish

- Technical Skills**
- Specialised in machine learning, astrostatistics, and large astrophysical datasets.
 - Python (NumPy, SciPy, JAX, PyTorch, Pandas), Git, LaTeX
 - Computational Astrophysics (CTA200, University of Toronto): received a final grade of 99%.

- Science Communication**
- Volunteer at Questacon – National Science and Technology Centre, Canberra, Australia (2022 – 2023)
 - Tutor, Freelance (2021 – 2023): Provided tutoring in Mathematics and Physics, in both French and English.
 - Tutor, Tutor Doctor ACT (Australian Capital Territory) (2021 – 2023): Provided tutoring for French, English, Mathematics and Physics.
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- OUTREACH**
- Astronomy Student Union Executive Member** (University of Toronto)
- President (2026/27 Academic Year)
 - VP External Affairs (2025/26 Academic Year)
 - Second Year Representative (2024/25 Academic Year)
- Volunteer, ASTROTOURS** (University of Toronto, 2024 – Present)
- Volunteer, Astro On Tap** (University of Toronto, 2024 – Present)
- Volunteer, Girls in STEM Workshop Event** (University of Toronto, May 2024)
- Member of Corporate Relations Team, Charitable Indulgence**
(University of Toronto, 2023/24 academic year)